

Test 1 · Algebra

Grade 8

40 minutes 100 marks 14 questions + bonus No calculator

- 1–3 Revision of the Grade 7 course
- 4–5 Algebraic expressions
- 6–8 Algebraic fraction. Cancelling fractions

Algebra 8, p. 3 · Stage 8 review

Algebra 8, §1 · Stage 9 · 2.1–2.2

Algebra 8, §2 · Stage 9 · 2.5

- Show your working — method carries marks.
- Whenever you cancel a fraction, state the permissible values of the variable.

What the paper weighs

TOPIC	MARKS	SHARE
Lessons 1–3 · Revision of the Grade 7 course	31	31%
Lessons 4–5 · Algebraic expressions	26	26%
Lessons 6–8 · Algebraic fraction. Cancelling fractions	43	43%

Cancelling carries the largest share because every question in Section C also exercises the factorising from lessons 1–3.

A Section A · Short answers

EASY 30 marks

Five marks each. One line of working is enough.

1 Simplify: $x^6 \cdot x^4$ [5]

ANSWER

x^{10}

WORKING

Same base, so the indices add: $6 + 4 = 10$.

2 Expand: $4(3x - 2)$ [5]

ANSWER

$12x - 8$

WORKING

Multiply both terms inside the bracket by 4.

3 Factorise: $10a - 15$ [5]

ANSWER

$5(2a - 3)$

WORKING

The common factor is 5.

- 4 State the permissible values of $\frac{1}{x-6}$ [5]

ANSWER

$$x \neq 6$$

WORKING

The denominator must not be zero: $x - 6 \neq 0$.

- 5 Find the value of $\frac{x+5}{x-2}$ when $x = 3$ [5]

ANSWER

$$8$$

WORKING

$$\frac{3+5}{3-2} = \frac{8}{1} = 8.$$

- 6 Cancel: $\frac{15x^3}{5x}$ [5]

ANSWER

$$3x^2, x \neq 0$$

WORKING

Divide numerator and denominator by $5x$.

B Section B · Eight marks each

MEDIUM

40 marks

Method marks are available even when the final answer is wrong.

- 7 Simplify: $(3a^3b^2)^2$ [8]

ANSWER

$$9a^6b^4$$

WORKING

Square every factor: $3^2 = 9$, $(a^3)^2 = a^6$, $(b^2)^2 = b^4$.

A common slip is $3a^6b^4$ — the 3 must be squared as well.

- 8 Factorise: $9x^2 - 16y^2$ [8]

ANSWER

$$(3x - 4y)(3x + 4y)$$

WORKING

A difference of two squares, since $9x^2 = (3x)^2$ and $16y^2 = (4y)^2$.

- 9 State the permissible values of $\frac{x+2}{x^2-25}$ [8]

ANSWER

$$x \neq 5, x \neq -5$$

WORKING

$$x^2 - 25 = (x - 5)(x + 5), \text{ which is zero at } x = 5 \text{ and at } x = -5.$$

- 10 For which value of x is $\frac{x-7}{x+1}$ equal to zero? [8]

ANSWER

$$x = 7$$

WORKING

A fraction is zero when its numerator is zero and its denominator is not: $x - 7 = 0$ gives $x = 7$, and that value is permissible because $x \neq -1$.

An answer of $x = -1$ shows the two conditions have been confused.

- 11 Cancel: $\frac{3x+12}{x^2-16}$ [8]

ANSWER

$$\frac{3}{x-4}, x \neq 4, x \neq -4$$

WORKING

Factorise first: $\frac{3(x+4)}{(x-4)(x+4)}$. The factor $(x+4)$ cancels.

C Section C · Ten marks each

HARD

30 marks

Factorise before you cancel, and state the values the variable may not take.

- 12 (a) Factorise completely: $x^3 - 9x$ [4]
(b) Hence cancel $\frac{x^3 - 9x}{x^2 + 3x}$ and state the permissible values. [6]

ANSWER

$$(a) x(x-3)(x+3) \quad (b) x-3, x \neq 0, x \neq -3$$

WORKING

(a) Take out x first, giving $x(x^2 - 9)$, then the difference of two squares gives $x(x-3)(x+3)$.

(b) The denominator factorises as $x(x+3)$, so $\frac{x(x-3)(x+3)}{x(x+3)} = x-3$. That denominator is zero at $x = 0$ and at $x = -3$, so both values are excluded.

“Completely” means going on until nothing factorises further; stopping at $x(x^2 - 9)$ loses marks.

- 13 Cancel $\frac{x^2 - 8x + 16}{x^2 - 16}$, state the permissible values, then find its value when $x = 9$. [10]

ANSWER

$$\frac{x-4}{x+4}, \quad x \neq 4, x \neq -4; \quad \text{at } x = 9 \text{ it is } \frac{5}{13}$$

WORKING

The numerator is a perfect square: $x^2 - 8x + 16 = (x - 4)^2$. The denominator is $(x - 4)(x + 4)$. One factor of $(x - 4)$ cancels and leaves $\frac{x - 4}{x + 4}$.

$$\text{At } x = 9: \frac{9 - 4}{9 + 4} = \frac{5}{13}.$$

- 14 (a) Cancel: $\frac{2x^2 - 2y^2}{3y - 3x}$ [7]
(b) Explain in one sentence why $\frac{x + 5}{5}$ cannot be cancelled. [3]

ANSWER

(a) $-\frac{2(x+y)}{3}, \quad x \neq y$ (b) because the 5 is a term, not a factor

WORKING

(a) $2x^2 - 2y^2 = 2(x - y)(x + y)$ and $3y - 3x = -3(x - y)$. The swap $y - x = -(x - y)$ is what makes the two factors match; cancelling $(x - y)$ leaves $-\frac{2(x + y)}{3}$.

(b) In $x + 5$ the 5 is a *term*, not a *factor*, and only a factor common to the whole numerator and the whole denominator may be cancelled.

Part (b) separates students who follow the rule from students who know why it holds.

Bonus

- ★ Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is BONUS +10 marks still recorded out of 100.

- 15 (a) Show that $\frac{m^2 - mn}{n^2 - mn} = -\frac{m}{n}$. [7]
(b) State the permissible values. [3]

ANSWER

(a) shown below (b) $n \neq 0, m \neq n$

WORKING

(a) $m^2 - mn = m(m - n)$. The denominator needs the swap: $n^2 - mn = n(n - m) = -n(m - n)$. So

$$\frac{m(m - n)}{-n(m - n)} = -\frac{m}{n}, \text{ since } (m - n) \text{ cancels.}$$

(b) The original denominator $n(n - m)$ is zero when $n = 0$ or when $n = m$, so both are excluded.

This is the sign swap from Q14(a) again, but with the minus hidden in the denominator rather than handed over. Full marks need the swap shown, not just asserted.